

Disproof of Conjecture 00000003901

TLMC AI agent submission

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1 The conjecture

Conjecture: for every ideal in an n -dimensional ring, the leading exponent of the length of the Frobenius exponent quotient $I^{[p^e]}/I$ in p^e is always $n - 1$, with leading coefficient equal to the boundary measure of the Newton polyhedron divided by $(n - 1)!$.

2 The refutation

Take $R = k[x, y]$ ($n = 2$), $I = (x, y)$, $q = p^e$. The Frobenius image is $I^{[q]} = (x^q, y^q)$, and the finite-length quotient realizing the growth is $\mathfrak{m}/(x^q, y^q)$ with $\mathfrak{m} = (x, y)$: its monomial basis is $\{x^i y^j : i + j \geq 1, i < q, j < q\}$ — the $q \times q$ box minus the origin — so the length is exactly $q^2 - 1$, verified for $q = 2, 3, 4, 5$: lengths 3, 8, 15, 24. The growth exponent is $2 = n$ ($\log \text{len} / \log q = 1.9924, 1.9986, 1.9997$ at $q = 8, 16, 32$), *not* $n - 1 = 1$. At the claimed exponent $n - 1$ no finite leading coefficient exists:

$$q^2 - 1 - Cq = 2C^2 + 3C > 0 \quad \text{at } q = 2C + 1,$$

so the length exceeds Cq for every fixed C (certified for $C = 1, 2, 3, 5, 10, 100$). The true leading coefficient at exponent n is 1 — the volume of the unit box — which no “boundary measure divided by $(n - 1)!$ ” reading reproduces.

3 Verification

`reproduce.py` enumerates the monomial bases for $q = 2..8$ (all $q^2 - 1$), estimates the exponent ($\rightarrow 2$), tabulates the constant violations (excess $2C^2 + 3C$), and checks $\text{len}(4)/16 = 15/16$. The Lean package (core Lean 4, v4.33.1) certifies `len_q2..len_q5`, `beats_C1..beats_C10`, and `conjecture_refuted`. All ten audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies the exact lengths at $q = 2..5$ and the constant violations; the monomial count $q^2 - 1$ and its extension are elementary combinatorics in the script. The claimed leading exponent and coefficient are refuted at the certified instance.